

ElaStick: A Stick-Type Controller with Elastic Material and IMU for Versatile Input and Feedback

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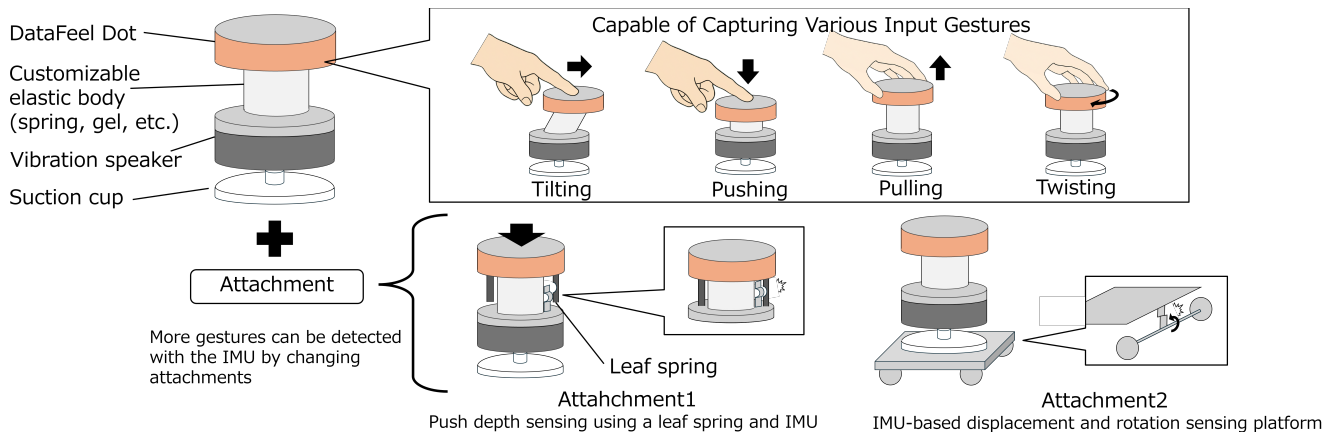


Figure 1: System Overview

Abstract

This project presents a modular stick-type controller toolkit based on the DataFeel Dots, integrating a 6-axis IMU and elastic materials. The toolkit supports sensing of common joystick and button inputs (e.g., pushing, tilting) as well as complex gestures like pulling and twisting. By enabling diverse gesture sensing with a single reconfigurable device, it offers a flexible platform for multimodal interaction. The system also provides four types of feedback—sound, vibration, light, and heat—enhancing user immersion. With interchangeable attachments, users can customize input by replacing elastic parts to adjust stiffness or adding extensions to detect parameters like push depth or bending. This flexibility supports creative applications in areas such as gaming, education, rehabilitation, and art. By unifying input and feedback in one toolkit, this work contributes to the development of a customizable toolkit for designing bidirectional controllers with integrated input sensing and multimodal feedback.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; *Interaction devices*; *Interactive systems and tools*.

Keywords

Modular Controller, Tangible Interaction, Elastic Materials, IMU Based Sensing, Multimodal Feedback, Toolkit

ACM Reference Format:

Nana Urakawa, Ryosuke Mizushima, and Shion Watanabe. 2025. ElaStick: A Stick-Type Controller with Elastic Material and IMU for Versatile Input and Feedback. In *The 38th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '25)*, September 28–October 01, 2025, Busan, Republic of Korea. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3746058.3758975>

1 Introduction

Tangible User Interfaces (TUIs) have been widely studied as a means of interacting with computers through physical manipulation of objects. While TUIs excel at supporting intuitive, hands-on interaction, challenges remain in establishing a generalizable framework that enables rich input and output expressions and dynamic multimodal feedback within a compact and simple form factor. The emergence of tools like DataFeel, which support multimodal feedback, along with advances in detecting user activities from high-level abstract information derived from IMUs, opens up new possibilities for developing flexible, bidirectional interface frameworks.

We propose a tangible and flexible controller that combines a 6-axis IMU from DataFeel Dot [1] and elastic material, allowing not only traditional operations like pushing and tilting but also more complex actions such as pulling and twisting to be captured as input. The elastic components can be customized with materials such as gels or springs of varying hardness, allowing users to adjust tactile and deformation characteristics as needed. In addition to capturing a variety of input gestures, the controller provides multimodal feedback through sound, vibration, light, and thermal stimuli. It is

also designed to support interchangeable attachments, allowing users to extend their input capabilities according to their needs.

2 RELATED WORK

Several studies and products have explored tangible interfaces that capture diverse deformation-based inputs using soft materials and embedded sensors. Twist Stick [2] is a stick-shaped device with a spring joint and tilt sensors at both ends, detecting bending, twisting, pulling, and pushing through relative angle measurement. Ring-Con [5] integrates a 6-axis IMU into a flexible ring, allowing gestures like squeezing, stretching, and tilting via pressure and motion sensing. BendID [4] uses a pressure-sensitive sheet to detect localized bending, enabling fine-grained recognition through machine learning. MaGEL [6] is a transparent soft device with embedded LEDs and phototransistors that classifies 13 deformation types based on light transmission patterns.

In the context of game controller design, thermal feedback has been explored as a means of conveying ambient information or enhancing emotional immersion. Peltier elements are employed by Comparing Thermal and Haptic Feedback Mechanisms for Game Controllers [3] to provide warm or cool sensations, conveying ambient or emotional cues. Beyond thermal feedback, other output modalities have also been explored to enrich interaction design in controllers. For example, GamesBond [7] simulates remote interpersonal interaction by synchronizing grip force and motion across devices. The DataFeel Devkit compactly integrates an IMU-based input sensor and multimodal output devices, such as thermal, vibration, and light, within a small 4 cm square module. Such device with multimodal output capability has the potential to be extended into a flexible controller by enhancing its input capabilities.

3 PROPOSED SYSTEM

3.1 System Overview

We propose a controller that provides physical manipulation and multimodal feedback using DataFeel Dot. This controller achieves diverse input methods and output expressions by combining a flexible elastic body and detachable attachments. The system overview is shown in Figure 1. The system components are provided below.

- **DataFeel Dot** detects operations using the built-in IMU and provides feedback through light, vibration, and temperature.
- **Elastic body** enables various interactions due to its flexible material. By altering the material properties, various tactile sensations can be created.
- **Vibration speaker** provides acoustic feedback in response to input.
- **Suction cup** can be attached to various surfaces, allowing for flexible placement adjustments.
- **Attachment** expands the controller’s operability and accuracy when mounted.

Users control the device by tilting, pushing/pulling, or twisting the mounted Dot. These actions are classified using a machine learning model based on the measurement data from the IMU, and corresponding feedback—light, vibration, and temperature—is provided through the DataFeel Dot. Acoustic feedback is delivered via a vibration speaker through the surface.

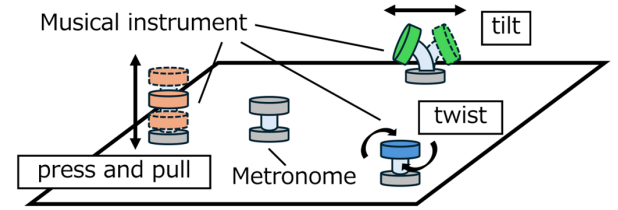


Figure 2: Application example: music session system

3.2 Application Example: Music Session System

As one of the applications of this controller, we propose a multi-user music session system (Figure 2). This system consists of four controllers equipped with DataFeel Dot: one serves as a metronome, and the remaining three function as musical instrument controllers.

- **The metronome controller** allows adjustment of the BPM through a twisting operation to the left or right, and the set BPM is presented as flashing light from the DataFeel Dot.
- **Each musical instrument controller** has a distinct instrument sound, with user gestures altering its timbre. Each controller features one of the following operation types: tilt, press and pull, or twist. Furthermore, different timbres are output depending on the direction of the operation. The frequency of three users input determines session excitement, shown as temperature feedback. The BPM of the metronome is also given as haptic feedback through vibration.

This system provides an experience where participants collaboratively create music using multiple controllers.

3.3 Extension Attachments

By adding attachments to the controller, we seek to expand the range of gestures that can be detected by the IMU. Below, we introduce two possible attachments we have considered. Their structures are shown in Figure 1.

- **Attachment 1** consists of a concave mold and a convex leaf spring mounted on the elastic body. When pressed, they interlock, providing a clear click and strong vibration. The IMU detects the impact generated by the press, enabling accurate recognition of the click event.
- **Attachment 2** adds mobility via rotating wheels, which generate vibrations through periodic plate collisions. The IMU captures these vibrations to estimate movement distance and speed, without external sensors.

Our module provides an idea for utilizing the IMU embedded in the DataFeel Dot to capture a wide range of input interactions. While the basic module alone can detect common controller interactions, various extension units can be attached to facilitate more diverse user behaviors for the IMU to sense, as described earlier.

By leveraging the Dot’s built-in feedback modalities, our module could serve as a foundation for designing controllers and tangible user interfaces.

Acknowledgments

We gratefully acknowledge the advice of Prof. Yuichi Itoh and Dr. Kouyou Otsu of Aoyama Gakuin University in shaping our idea.

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